J1
GNSS: Foxeer M10Q-250 / Beitian BE-220
+5V 1
GND 2
GNSS_TXD 3
GNSS_RXD 4
SCL 5
SDA 6

1 VCC 5V 2 GND 3 TX -> IO4 4 RX <- IO5
5 SCL / 6 SDA: compass (Foxeer only, 0x0D). BE-220 has pins 1-4.

J3
SSD1306 0.96" OLED I2C 0x3C
GND 1
+3V3 2
SCL 3
SDA 4

1 GND 2 VCC 3V3 3 SCL <- IO9 4 SDA <> IO8
Module pull-ups on SDA/SCL; bus at 100 kHz.

+5V
C2
1000u 35V low-ESR
GND

J2
SIM800C GPRS board
+5V 1
GND 2
GSM_TXD 3
GSM_RXD 4
GSM_PWRKEY 5
+3V3 6

1 5V in 2 GND 3 TXD -> IO18 4 RXD <- IO17 5 PWRKEY <- IO7 6 V_TTL = 3V3
Board without level shifter: 1.8k/3.3k divider on RXD (chip logic is 2.8 V).
2 A transmit bursts: C2 right at pins 1-2, short thick wires.

SW1
Button (BOOT-style, active low)
BTN
GND

Internal pull-up on IO2; GPIO interrupt on both edges.

U1
ESP32-S3-WROOM-1 (DevKitC-1)
3V3_DEVKIT
EN 3
3V3 2
IO0 27
IO1 39
IO2 38
IO3 15
IO4 4
IO5 5
IO6 6
IO7 7
IO8 12
IO9 17
IO10 18
IO11 19
IO12 20
IO13 21
IO14 22
IO15 8
IO16 9
TXD0 37
RXD0 36
IO17 10
IO18 11
USB_D- 13
USB_D+ 14
IO21 23
IO35 28
IO36 29
IO37 30
IO38 31
IO39 32
IO40 33
IO41 34
IO42 35
IO45 26
IO46 16
IO47 24
IO48 25
GND 1
PSRAM
BATT_SENSE
BTN
GNSS_TXD
GNSS_RXD
ACC_INT1
GSM_PWRKEY
SDA
SCL
SD_CS
SD_MOSI
SD_SCK
SD_MISO
GSM_RXD
GSM_TXD
GND

J4
MicroSD SPI module
GND 1
+3V3 2
SD_MISO 3
SD_MOSI 4
SD_SCK 5
SD_CS 6

1 GND 2 VCC 3V3 (or 5V if the module has a level shifter)
3 MISO -> IO13 4 MOSI <- IO11 5 SCK <- IO12 6 CS <- IO10

J6
BMA400 CJMCU-400 (not fitted yet)
+3V3 1
GND 2
SDA 3
SCL 4
ACC_INT1 5
6

1 VCC 3V3 2 GND 3 SDA 4 SCL 5 INT1 -> IO6 (RTC GPIO, deep-sleep wake) 6 INT2 n/c
I2C 0x14 (0x15 with SDO high).

J5
Battery Shield V8, 2x18650
+5V 1
+3V3 2
GND 3
+BATT 4

1 5V out (<= 2.2 A cont.) 2 3V out (1 A; measure: 3.0 or 3.3 V) 3 GND 4 B+ (cell)
Feeds: +5V -> devkit 5V pin, GNSS, SIM800C board. +3V3 -> OLED, SD, BMA400.

+BATT
R1
100k
BATT_SENSE
R2
100k
GND

Cell 4.2 V max -> 2.1 V at IO1 (ADC1_CH0, 12 dB).

DevKitC-1: 5V pin <- +5V rail (onboard LDO makes 3V3_DEVKIT).
USB, EN, BOOT, JTAG and the onboard RGB LED (IO38/IO48) stay on the devkit.
SuperMini build: GSM on IO43/IO44 instead of IO17/IO18.

+5V PWR_FLAG
+3V3 PWR_FLAG
+BATT PWR_FLAG
GND PWR_FLAG
PWR_FLAG
3V3_DEVKIT

PWR_FLAG: the rails are sourced by the battery shield (J5); 3V3_DEVKIT by the devkit's LDO.

Breakout modules drawn as pin headers; see docs/HARDWARE.md
ESP32-S3 DevKitC-1 development build; pins per include/config.hpp
Maksym Kutyshenko

Sheet: /
File: gps_tracker.kicad_sch

Title: GPS Tracker

Size: A3	Date: 2026-10-09	**Rev: 1**
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